Starter/Generator – General Data

1. General

The starter/generator is mounted on the upper rear face of the engine accessory gear case with a quick-disconnect kit. The kit consists of a drive end pad and a clamp. The pad is secured to the accessory gear case with four studs, washers and nuts, and the clamp is used to attach the starter/generator to the drive end pad. The starter generator is a wet spline drive shaft with an O-ring seal installed on the shaft. The starter/generator is cooled by air drawn from the engine intake area forward of the engine rear fire seal. The cooling air is ducted to the rear of the starter/generator to a duct assembly which is clamped to the starter/generator. The cooling air flow travels forward through the starter/generator, augmented by an integral four-bladed fan, and is exhausted through a screen at the drive end. For further maintenance details refer to the starter-generator Component Maintenance Manual (CMM). Maintenance of the LRU beyond that described below should be done at an accredited maintenance depot.

2. Removal/Installation

Note For unscheduled starter/generator removal, refer to Pratt and Whitney Engine Maintenance Manual, Chapter 72-00-00, Unscheduled Inspection, Starter-Generator Replacement.

Refer to Figure 1 for details.

A. Remove Starter/Generator

1) Ensure electrical power is OFF.

2) Gain access to starter/generator through engine nacelle.

a) Remove clamp securing cooling air duct hose to duct assembly. Disengage cooling air duct and remove air duct.

b) Loosen clamp securing duct assembly to rear of starter/generator fan cover.

c) Loosen clamp securing duct assembly to rear of starter/generator fan cover and remove duct assembly.

d) Remove four screws securing fan cover and carefully remove fan cover.

3) Remove electrical leads from terminal block.

CAUTION STARTER/GENERATOR COOLING FAN BLADES MAY BE DAMAGED OR BENT ON REMOVAL.

4) Support starter/generator and remove retaining clamp from drive end pad.

5) Slide starter/generator rearward to disengage drive. Remove starter/generator.

6) Remove felt pad from starter/generator.

Note The starter/generator drive end pad need only be removed when changing an engine. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

B. Install Starter/Generator

1) Check starter/generator drive gearshaft splines for wear in accordance with procedures in the CMM.

2) Ensure electrical power is OFF.

3) Position felt pad around rear of starter/generator.

4) Install O-ring seal on starter generator shaft.

CAUTION **STARTER/GENERATOR MAY BE DAMAGED OR BENT ON INSTALL.**

STARTER/GENERATOR DRIVE GEAR CARBON SEAL CAN BE EASILY DAMAGED IF NOT CAREFUL ON INSTALLATION.

5) Carefully position starter/generator drive end to drive end pad, with terminal block at 12 o'clock position. Engage drive.

6) Secure starter/generator to drive end pad with clamp. Torque clamp to 50 pound–inches.

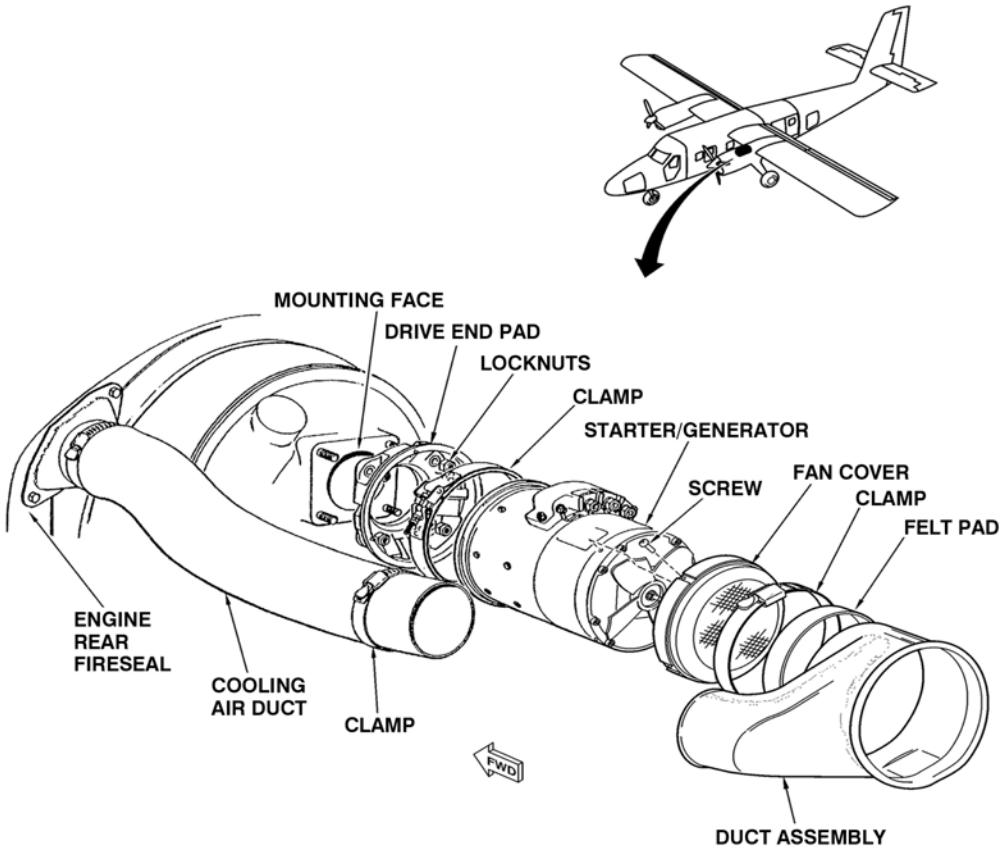
7) Secure electrical leads to terminal block.

8) Install fan cover with four screws, install air duct assembly.

9) Ensure duct assembly is against firewall. Install duct assembly retaining clamp and torque to 25 pound–inches.

10) Install cooling air duct hose to duct assembly and secure with clamp.

11) Function test starter/generator. (Refer to 24-30-00, DC Generation – General Data).



Starter/Generator Removal and Installation
Figure 1

3. Adjustment/Test

   A. Bench Test Starter/Generator

      Bench testing of the LRU should be done in accordance with the CMM and at an accredited maintenance depot.

   B. Adjust Brush Seating and Brush Run-in Starter/Generator

      Refer to Goodrich Corporation Standard Practices Document SPD 1006.

   C. Flashing the Generator Field (Bench Test)

      Do in accordance with the CMM and at an accredited maintenance depot.

4. Inspection/Check

   A. Check Brush Wear Starter/Generator

      1) Gain access to starter/generator.

      2) Inspect brushes and brush holders in accordance with the CMM.

      3) Replace brushes if remaining allowable wear is liable to be exceeded before next inspection.

         Note To facilitate brush seating adjustment and brush run-in (refer to Para 3.B) when brushes are replaced, the starter/generator should be removed (refer to Para 2.A).

   B. Check Brush Spring Tension

      Check brush spring tension in accordance with the CMM.

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